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High-Performance Monolithic 3D Integrated Complementary Inverters Based on Monolayer n-MoS_2 and p-WSe_2

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Herein, the structure of integrated M3D inverters are successfully demonstrated where a chemical vapor deposition (CVD) synthesized monolayer WSe_2 p-type nanosheet FET is vertically integrated on top of CVD synthesized monolayer MoS_2 n-type film FET arrays (2.5 × 2.5 cm) by semiconductor industry techniques, such as transfer, e-beam evaporation (EBV), and plasma etching processes. A low temperature (below 250 °C) is employed to protect the WSe_2 and MoS_2 channel materials from thermal decomposition during the whole fabrication process. The MoS_2 NMOS and WSe_2 PMOS device fabricated show an on/off current ratio exceeding 10^6 and the integrated M3D inverters indicate an average voltage gain of ≈9 at V_{DD} = 2 V. In addition, the integrated M3D inverter demonstrates an ultra-low power consumption of 0.112 nW at a V_{DD} of 1 V. Statistical analysis of the fabricated inverters devices shows their high reliability, rendering them suitable for large-area applications. The successful demonstration of M3D inverters based on large-scale 2D monolayer TMDs indicate their high potential for advancing the application of 2D TMDs in future integrated circuits.

1. Introduction

Metal-oxide-semiconductor (MOS) field effect transistors (FETs) are the fundamental building blocks for contemporary very-large-scale integrated circuits. As silicon-based transistors reach the sub-10 nm technology node, numerous challenges, such as short the channel effect (SCE), and drain-induced barrier lowering (DIBL) have emerged in conventional MOS FETs.[1] Hence, new device structures and alternative channel materials have been proposed.[2] For example, FinFETs,[3] gate-all-around (GAA) structures,[4] stacked inverters,[5] and multi-bridge channel FET,[6] have been demonstrated to enhance the gate controllability of the channel and mitigate the SCE and DIBL effects. Especially, stacked inverters are a highly promising candidate for ultra-scaling transistors due to their small footprint while maintaining high performance. However, conventional stacked inverters fabricated by 3D semiconductors, such as silicon,[7] germanium,[8] conventional III-V compounds,[9] and IGZO,[10] have shown undesired performance when the thickness is scaled down to sub-5 nm. Meanwhile, 2D semiconductors channel materials (SCM), such as transition metal dichalcogenides (TMDS), have emerged as promising alternatives to 3D semiconductors

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because of their dangling-bond-free surfaces and low mobility variation with decreasing thickness.[11]

3D integrated circuit (3D IC) technology has emerged as an alternative approach for addressing the trade-offs among performance, density, and power dissipation via the stacking of multiple transistor layers. Two primary types of 3D IC technologies exist. The first type is through-silicon vias (TSV)-based 3DIC, in which devices are fabricated separately on different wafers and then integrated via a bonding process. TSVs have larger critical dimensions than the standard back-end-of-line (BEOL) vias for planar integrated circuits. Owing to their large size, TSVs are more suitable for block-scale signal transport rather than for fine-grained integration. The second type is the monolithic-3DIC (M3D), in which devices are sequentially fabricated on the same wafer. Each layer of the device comprises an inter-layer dielectric (ILD) and multiple monolithic intertier vias (MIVs). These MIVs have dimensions similar to conventional planar IC vias, thus allowing for a higher integration density and signal transport at the device scale.[12] The M3D approach enables a higher integration density and more efficient signal routing to be achieved in the IC. In M3D technology, multiple layers of transistors and interconnects can be stacked, thus enabling the integration of different functions on a single chip.[12] In addition, M3D technology consumes significantly less power than conventional planar ICs.[13] By stacking transistors vertically, shorter interconnects are achieved, thus reducing the resistance and capacitance, which consequently reduces power consumption. By leveraging this technique, the distinctive features of 2D materials can be harnessed effectively, thus enabling their incorporation into advanced 3D IC architectures. Recently, M3D inverters based on 2D TMDs have received significant interest owing to their ultrathin bodies and excellent electrical performance. Huang et al. demonstrated an M3D integrated complementary FET (n-MoS_2−p-WSe_2) for low-power ICs. The M3D integrated complementary FET showed an ultra-high on-current (≈594 μA μm^{-1}). Furthermore, they were the first to report experimental results pertaining to 4T SRAM and 16T half-adder circuit units.[14] Xia et al. demonstrated the ability to switch light signals to electrical digital signals by an integrated inverter (n-MoS_2/p-$MoTe_2$), that exhibited a voltage gain of 7 V/V at V_{DD} = 3 V.[15] Hu et al. demonstrated a wafer-scale of integrated FETs (n-MoS_2−p-WSe_2) with a high gain of 10 V/V at V_{DD} = 3 V.[16] Additionally, an M3D inverter structure combining back-end compatible 2D TMDs on a silicon-based FET was demonstrated.[17] For example, the integrated n-type MoS_2 triangle domains on a p-Si FinFET with a fin width of 20 nm showed a large gain of ≈38.[17a] These innovative studies demonstrate the great potential of TMDs-based integrated inverters for use in high-performance devices. However, fabricating monolayer TMDs-based M3D inverters on a large scale remains challenge because of the fragility of monolayer TMDs and the multi-step fabrication process involved. Monolayer TMDs are sensitive to environmental factors and degrade easily during fabrication. Additionally, the sequential integration of device tiers and the precise alignment of different layers pose challenges in achieving large-scale production.

Herein, M3D integrated inverters based on monolayer *n*−type MoS_2 films and *p*−type WSe_2 flakes are presented. The 1st layer of MoS_2 top-gate FET arrays was defined by standard back-end-of-line (BEOL) compatible semiconductor technology. The ILD deposition and monolithic interior vias (MIVs) are deposited by plasma-enhanced chemical vapor deposition (PECVD) and physical vapor deposition (PVD), respectively. Subsequently, local back-gated WSe_2 FETs are defined as the 2nd layer of the device. To prevent the decomposition of TMDs, the temperature of the M3D integration was maintained below 250 °C. Both our MoS_2 NMOS, and WSe_2 PMOS device show an on/off current ratio exceeding 10^6 and the fabricated M3D inverter exhibited a maximum voltage gain of 9.5 at V_{DD} = 2 V. Statistical data of the M3D inverter devices indicated excellent uniformity, which render them suitable for large-area applications.

2. Results and Discussion

Figure 1a shows a schematic illustration of the proposed M3D integrated inverters, where MoS_2 NMOS, and WSe_2 PMOS M3D were vertically integrated. In this work, large-area *n*-type MoS_2 film and *p*-type WSe_2 flakes were prepared by CVD method. First, MoS_2 film was grown on a single-crystal sapphire substrate using low-pressure chemical vapor deposition (LPCVD) (see details in the Experimental section). Detailed characterizations of MoS_2 film, including optical microscopic (OM) images, photoluminescence (PL), and Raman spectroscopy characterizations, are shown in Supporting Information. The OM images of the LPCVD-grown MoS_2 monolayer are shown in Figure S1 (Supporting Information). With a short growth time, antiparallel domains with a typical size of ≈20 μm were obtained (Figure S1a, Supporting Information). By extending the growth time, MoS_2 film was fully coalesced with the domains as shown in Figures S1b and S1d (Supporting Information). Previous studies showed that the film coalesced by high-oriented MoS_2 showed excellent electrical performance.[18] Raman spectra were performed to check the grown MoS_2 monolayer film. As shown in Figure S2a (Supporting Information), two typical Raman peaks of A_{1g}(≈385 cm^{-1}) and E_{2g}(≈404 cm^{-1}) were found in all randomly selected points of the as-grown MoS_2 monolayer film. The distance between the two peaks was ≈19 cm^{-1}, thus indicating that the as-grown MoS_2 film was a monolayer. In addition, only one PL peak at 1.87 eV has a narrow full width at a half-maximum of 57 meV (Figure S2b, Supporting Information). These results indicated that the prepared MoS_2 film was of high-quality. Then, the prepared MoS_2 film was transferred onto the target substrate via the KOH solution-assisted method (see details in the experimental sections and Figure S3, Supporting Information). After transferring the MoS_2, the 1st layer of n-type MoS_2 FET arrays (2.5 x 2.5 cm) was defined by the deposition of the top gate dielectric materials, local top-gate electrodes, and source−drain electrodes. Similarly, the 2nd layer of *p*-type WSe_2 FETs was defined by the deposition of the bottom gate dielectric materials, bottom gate electrodes, and source−drain electrodes. Finally, the 1st layer and 2nd layer were connected by metal deposition in accordance with the inverter design.

Figure 1b–j presents detailed illustrations of crucial processes for fabricating the M3D integrated inverters, such as inter-layer deposition, contact region etching, and the creation of monolithic intertier vias (MIVs) interconnection. After transferring the MoS_2 film onto a SiO_2/Si substrate, 3 nm SiO_2 and 5 nm HfO_2 films were deposited by EBV and plasma-enhanced atomic layer deposition (PEALD), respectively (Figure 1b). The deposition of both

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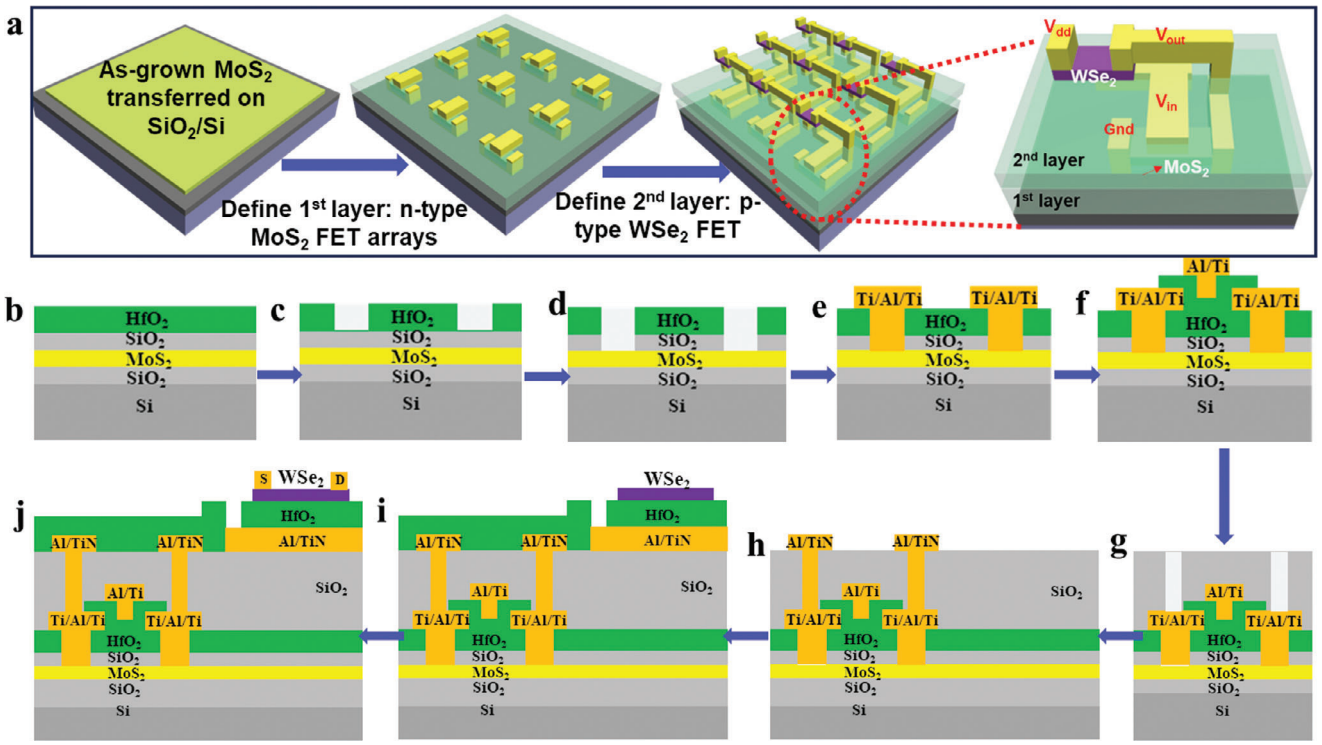

Figure 1. Processes of M3D integrated inverter a) Schematic illustration of the preparation of M3D integrated inverter. b) After transferring the 1L MoS_2 continuous film onto the target substrate, 3 nm-thick SiO_2 and 5 nm-thick HfO_2 films were deposited. c) Contact regions on HfO_2 defined by dry etching. d) Contact regions on SiO_2 defined via BOE wet etching. e) Source/drain metallization by Ti/Al/Ti electron beam deposition f) Top gate dielectric deposition of 10 nm-thick HfO_2 by ALD and top gate metallization by Al/Ti electron beam deposition g) 300 nm-thick SiO_2 ILD deposition by PECVD under 250 °C and via pattering. h) Interconnect deposition with Al/TiN by PVD and patterning. i) Local back gate deposition and patterning followed by high-κ dielectric deposition and WSe_2 transfer j) WSe_2 PFET source/drain patterning followed by connection with the bottom layer MoS_2 NFET, which formed a 3D integrated inverter.

SiO_2 and HfO_2 serves to prevent direct exposure of the MoS_2 film to the e-beam resist, thereby protecting the device performance from potential degradation.[19] Subsequently, isolated arrays of MoS_2 channels were created as active regions by O_2 plasma dry etching. Owing to the complete coverage of the MoS_2 film by SiO_2 and HfO_2 dielectric layers, a two-step etching process is required to create contact windows for the deposition of source/drain metals. Hence, the HfO_2 layer was etched by BCl_3 plasma treatment, whereas the SiO_2 layer was etched using a buffered oxide etchant (BOE) solution (Figure 1c,d). Next, Ti/Al/Ti (5/30/5 nm) source and drain electrodes were fabricated by EBV (Figure 1e). After metalizing the source and drain, an additional film of HfO_2 with a thickness of 10 nm was deposited as the top gate dielectric using PEALD. Subsequently, a similar metallization process was carried out to fabricate the top gate electrode (Al/Ti 30/5 nm) (Figure 1f). Upon completing the first layer of the MoS_2 FETs array, we deposited a 300 nm-thick SiO_2 layer by PECVD at the temperature of 250 °C (Figure 1g). This SiO_2 layer serves as an ILD, serving to prevent current leakage resulting from specific electrical coupling effects between adjacent layers.[20] The deposition of the ILD at low temperature is crucial for preventing the degradation of the bottom-tier MoS_2 devices during M3D IC fabrication.[20b] As illustrated in **Figure 2**g,h, after depositing the ILD, contact holes were created by dry etching the SiO_2 layer and were subsequently filled with 300 nm of Al and 10 nm of Ti by EBV. These metal layers serve as the MIVs between the second and first layers. For the 2nd layer of the device, we define local back-gated WSe_2 PMOS with CVD-grown flakes as its single crystallinity offers high device performance. Single crystal WSe_2 flakes were prepared by CVD method (details are shown in the Experimental Section). As shown in Figure S4 (Supporting Information), the Raman spectrum showed an E^1_{2g} peak at 243 cm^{-1} and no peaks at 304–307 cm^{-1}, thereby indicating that the as-grown WSe_2 is a monolayer.[21] Additionally, the PL spectrum only shows a sharp peak at ≈1.68 eV, thus confirming the high quality of the as-grown WSe_2 monolayer. The fabrication processes of the local back-gated WSe_2 PMOS are illustrated in Figure 1i,j. The local back gate arrays were connected to the top gate of the bottom layer MoS_2 FETs using 30 nm-thick Al and 5 nm-thick Ti films, which served as the input terminals (V_{in}) for our M3D logic inverters. After transferring the WSe_2 flakes, OM was performed to locate the WSe_2 flakes that intersected one of the local back gate electrodes. Subsequently, e-beam lithography was employed to define the source/drain regions, followed by the deposition of Ti/Al/Ti (5/30/5 nm) electrodes using EBV. All the M3D integration processes of the MoS_2 NMOS and WSe_2 PMOS devices were fabricated at temperatures below 250 °C, rendering them compatible with BEOL processes used in modern semiconductor technologies. This compatibility indicates the potential for the industrialization of transistors fabricated using 2D materials.

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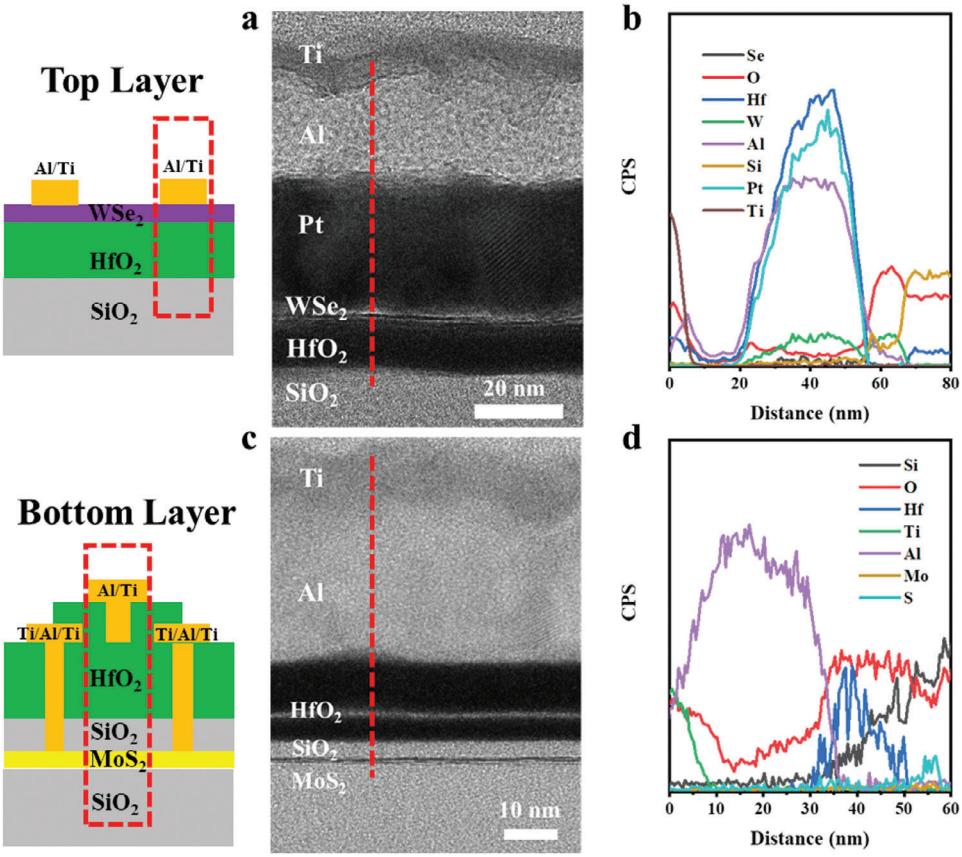

Figure 2. a,c) Cross-sectional high-resolution transmission electron microscope images of the 1st and the 2nd layers, respectively. b,d) Corresponding energy-dispersive X-ray spectroscopy elemental line scan profiles across the red line in a) and c), respectively.

To confirm that the integrated heterointerfaces of M3D was not damaged, we performed the cross-sectional imaging using the high-resolution transmission electron microscopy (HR-TEM) and energy dispersive X-ray Spectroscopy (EDX) element line scans. As shown in Figure 2a,c, the TEM images exhibited sharp and clear interfaces for all the integrated layers. Meanwhile, the EDX line analysis results in Figure 2b,d confirmed the uniform distribution of all elements without any interdiffusion. Furthermore, the MoS_2 and WSe_2 channels were effectively encapsulated by dielectric layers, thus providing excellent protection and isolation. Previous studies demonstrated that direct atomic layer deposition (ALD) on 2D materials typical results in the formation of porous films or island growth due to random nucleation on defects or impurities present on the 2D surface.[22] To address this issue, numerous methods have been examined to achieve high-quality growth of high-κ dielectrics on 2D material surfaces. These methods include the implementation of a buffer layer,[23] UV-ozone treatment,[24] and plasma treatment.[25] In the top-gate MoS_2 FETs, a SiO_2 layer was first deposited as a buffer layer before the ALD of HfO_2. The SiO_2 layer served as a protective layer against damage during the ALD and as a seed layer that facilitates the conformal and pinhole-free deposition of the HfO_2 film.

The electrical performance of 1st layer of FETs was first examined prior M3D fabrication. **Figure 3a** shows an optical image of the bottom layer MoS_2 top gate arrays, fabricated using a CVD grown large-scale 1L MoS_2 continuous film. Figure 3b,c shows the I_{DS}–V_{GS} characteristics of 1st top gated MoS_2 FET at drain voltages of 0.1 and 1 V, respectively. At $V_{DS} = 0.1$ V, the on-current (I_{on}) was $\approx 10^{-7}$ A and the maximum on/off ratio was 10^5 whereas the off current decreased to ≈1 ppA. When V_{DS} increased to 1 V, the maximum I_{on} was 4.5 μA and the on/off ratio reached 4×10^6. The subthreshold swing (S.S) value was determined as follows:

$$S.S. = \frac{\partial V_{GS}}{\partial(\log_{10} I_{DS})} \tag{1}$$

the S.S. value was calculated to be 314 mV dec^{-1} at $V_{DS} = 1$ V. Furthermore, we calculated the equivalent oxide thickness (EOT) of our top-gated MoS_2 FETs, which is the thickness of SiO_2 that provides the same capacitance, as follows:

$$EOT = \frac{\varepsilon_0 \varepsilon_{SiO_2}}{C_i} \tag{2}$$

where C_i denotes the capacitance of the gate dielectric. The EOT of the top-gated MoS_2 FET was calculated to be 6.65 nm. The field effect mobility (μ) was estimated from the transfer curve by as follows:

$$\mu = \frac{L}{W} \frac{g_m}{C_{ox} V_{DS}} = \frac{L}{W} \frac{1}{C_{ox} V_{DS}} \frac{\partial I_{DS}}{\partial V_{GS}} \tag{3}$$

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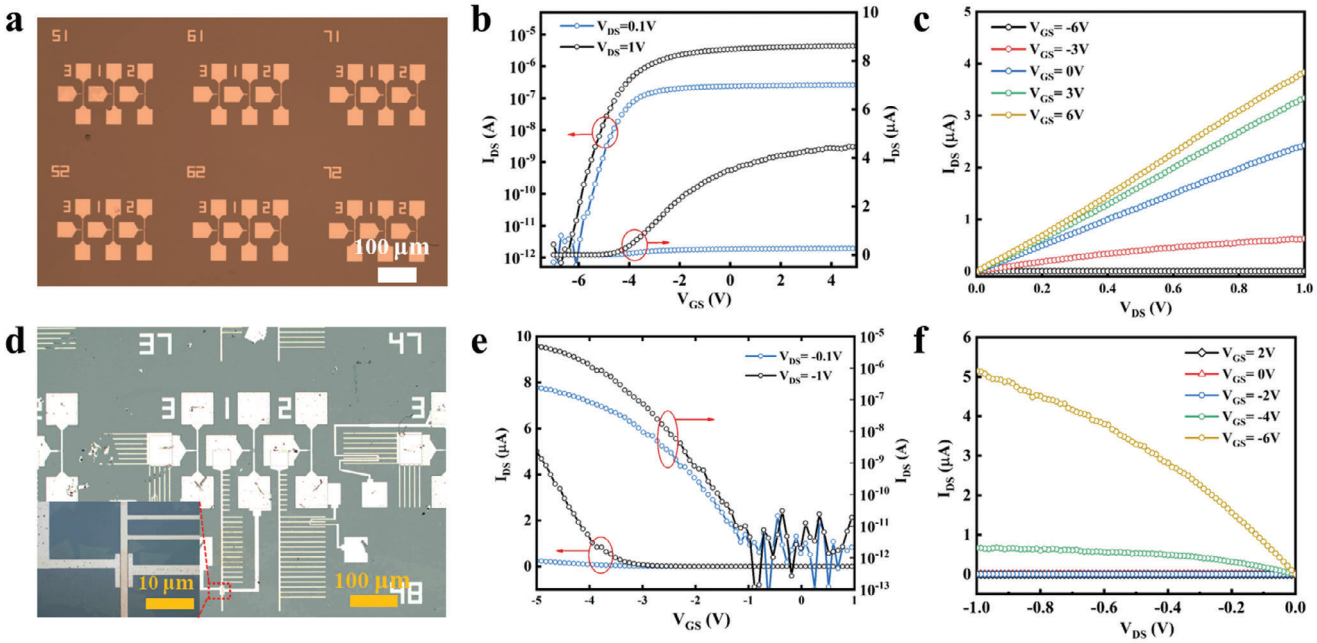

Figure 3. a) Optical image of bottom layer MoS_2 top gate NFET arrays. b,c) Transfer and output characteristics of MoS_2 NMOS before ILD deposition. d) An optical image of top layer WSe_2 PFET arrays. e,f) Transfer and output characteristics of WSe_2 PMOS.

where W and L represent the channel width and length of our device, respectively. Here, the term g_m is the transconductance, which is the first derivative of the drain current in the linear region. The W/L of our fabricated device was 1 µm/3 µm. The calculated mobility value was 6.45 cm^2V^{-1} s^{-1}. Figure S5a (Supporting Information) shows the transfer curves of 33 top-gated MoS_2 FETs devices on the same chip at V_{DS} = 1 V. The variation between reference voltages (selected based on the V_{GS} at I_{DS} = 10^{-9} A) was 2 V. Additionally, as shown in Figure S6 (a–c) (Supporting Information), the device-to-device variations of MoS_2 FET arrays were also investigated in terms of maximum drain current, S.S, and field effect mobility, respectively. Excellent statistical results indicated good uniformity across a large-scale array, thus conforming the effectiveness and reliability of our fabrication method. Figure 3d shows an optical image of top layer WSe_2 top gate PFETs. Previous studies have demonstrated that the contact metals utilized in WSe_2 FETs not only determine the injection efficiency of holes but also significantly affect the characteristics of carrier transport within the device.[26] High-work function metals, such as palladium (Pd) and platinum (Pt), are suitable for WSe_2 FETs.[27] Figure 3e,f showed I_{DS}–V_{GS} measurements of the 2nd local back-gated WSe_2 PMOS device at different drain voltages of 0.1 and 1 V, respectively. The device shows an on/off current ratio exceeding 10^6 with an on-state current exceeding 10^{-6} A (I_{ON} ≈5 µA) and an off-state current of ≈10^{-12} A (I_{OFF} ≈5 pA) at V_{DS} = 1 V. The threshold voltage (V_{th}) of this particular device was −3.52 V, as determined using the linear extrapolation method. Furthermore, the calculated S.S. value was ≈325 mV dec^{-1} and the field-effect mobility was ≈1.3 cm^2 V^{-1} s^{-1}. Figure S5b (Supporting Information) shows the transfer curve of 10 local back-gated WSe_2 FETs on a single chip. The ambipolar transport characteristic of these devices are reflected by a hole current exceeding 10^{-6} A and an electron current around of 10^{-8}≈10^{-7} A (50–100 nA).[28] In addition, the distribution of 25 local back-gated WSe_2 FETs at V_{DS} = -1 V in terms of maximum drain current, S.S., and field effect mobility is shown in Figure S7 (a–c) (Supporting Information), respectively.

2.1. Upon Confirming the Individual Performance of the 1st Top-Gated MoS_2 NMOS and the 2nd

Local back-gated WSe_2 PMOS devices, we evaluated the performance after the fabricated M3D integrated inverters. **Figure 4a** shows the transfer characteristics of MoS_2 and WSe_2 in the integrated M3D inverter device. The electrical performance of MoS_2 after M3D integration was similar to that before integration because of the low-temperature processes used for the constructing of the upper-tier components. The upper tier WSe_2 PMOS and bottom tier MoS_2 NMOS devices exhibit similar on-state performance where the on/off current ratio was of the same order of magnitude. Figure 4b presents the voltage transfer characteristic curve of theM3D integrated inverters. The voltage gain was calculated as follows:

$$Gain = \left|\frac{dV_{out}}{dV_{in}}\right| \tag{4}$$

the gain is over 1 for the applied V_{DD}, which satisfied the operating condition of the inverter. The maximum voltage gain of the MoS_2 NMOS/WSe_2 PMOS M3D inverter was calculated to be 9.55 V/V at V_{DD} = 2 V. The device exhibited an output voltage swing in comparison with the supply voltage. This behavior may be caused by the accumulation of trap charges at the interface of $MoS_2/SiO_2/HfO_2$ during fabrication. The presence of these trapped charges can introduce additional electrons

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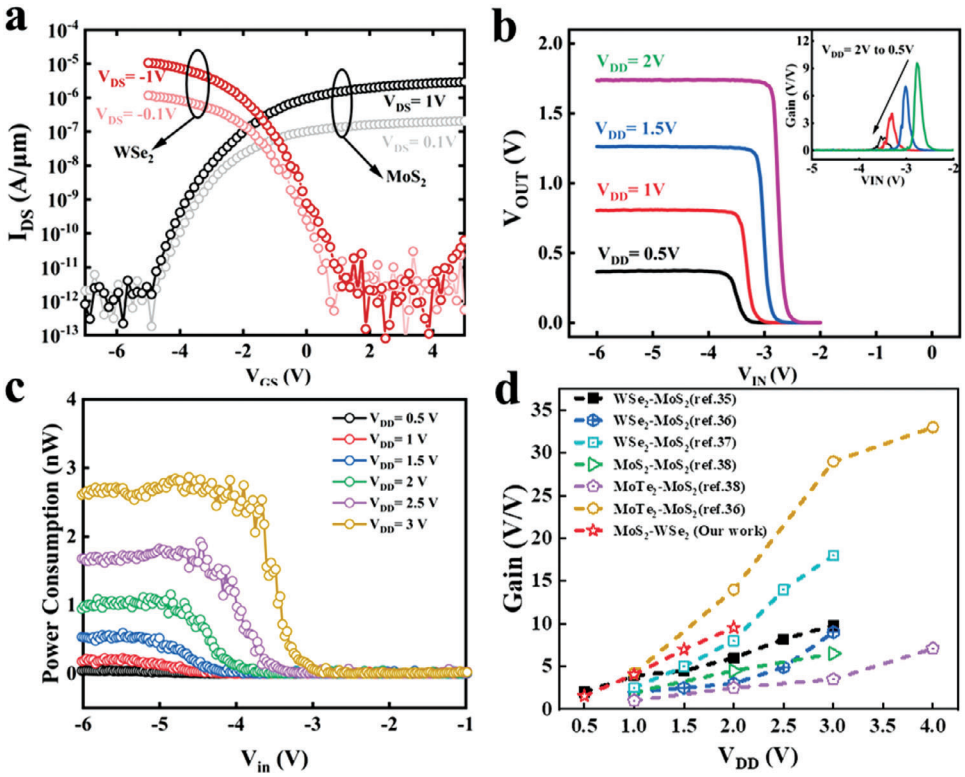

Figure 4. a) Transfer characteristics of MoS_2 and WSe_2 from stacked M3D inverter devices. b) Voltage transfer characteristics of the M3D integrated inverter under power supply voltage (V_{DD}) from 0.5 to 2 V in 0.5 V steps. Inset shows corresponding voltage gain at different V_{DD}. c) Power consumption characteristics of the inverter (b) at various V_{DD}. d) Benchmark of gain versus V_{DD} for the 2D TMD-based inverter.

into the channel, thereby increasing the off-state current.[29] Because the negative threshold voltage of both the MoS_2 NMOS and WSe_2 PMOS transistors, the transition voltage of our M3D inverter was below 0 V. Figure S8 (Supporting Information) shows the distribution of the voltage gain for 25 MoS_2 NMOS/WSe_2 PMOS M3D inverters at a supply voltage (V_{DD}) of 1 V. Clearly, 22 of the 25 inverters analyzed exhibited functional behavior, as they indicated a voltage gain >1. The limited voltage gain of our MoS_2/WSe_2 M3D inverters at lower supply voltages is attributed to two factors: the threshold voltage mismatch and the ambipolar transfer characteristic of the WSe_2 PMOS transistor at V_{GS} >0 V. Figure 4c shows the power consumption ($V_{DD} \times I_{DD}$) characteristics of this device. The minimum power consumption was as low as 0.037 and 0.112 nW at V_{DD} of 0.5 and 1 V, respectively. In addition, we have summarized the power consumption of these relevant studies and presented the results in Table S1 (Supporting Information). In Figure 4d, we present a performance comparison of our as-prepared M3D integrated inverters in terms of voltage gain with other structures of similar architectures based on 2D TMDs-based materials. In particular, These other structures included MoS_2/WSe_2 M3D inverter with exfoliated flakes,[30] MoS_2/WSe_2 integrated inverters with CVD-grown films,[31] MoS_2-based depletion-load 3D inverter with CVD-grown films,[32] and MoS_2/$MoTe_2$ integrated inverters with CVD-grown films.[32,33] Clearly, our inverter achieves voltage gain comparable to those of other structures at low supply voltages (V_{DD} = 0.5–1.5 V). Although we successfully demonstrated a full 2D-FETs M3D inverter, its performance can be further improved. The modulation of threshold voltage in our 2D-FETs must be improved to achieve a higher voltage gain for the inverter and to enable the realization of more advanced circuits, such as SRAM. In this context, interface dielectric engineering or dual gate modulation may be a possible method to enhance the control over the threshold voltage of our devices. Additional measurements such as capacitance-voltage (C–V) and low-frequency noise analysis can aid in determining the quality of the gate dielectric and interface, as well as provide more accurate trap density extraction. Moreover, to render our WSe_2 p-channel FETs fabrication process more stable and suitable for industrial applications, a more compatible approach must be developed because the current local back gate configuration may not align well with modern IC technology requirements. Finally, more advanced 3DIC configurations such as integrated inverters and GAA NS[34] should be further explored.

3. Conclusion

In summary, we demonstrated a monolithic 3D integrated logic inverter utilizing a large-scale top-gated MoS_2 FET array as the first layer and local back-gated WSe_2 FETs as the second layer. To prevent device degradation, we employed a low thermal budget (250 °C) BEOL-compatible process. Both the MoS_2 NMOS and WSe_2 PMOS devices exhibited excellent on/off current ratios exceeding ≈10^6. The fabricated M3D integrated inverter achieved a voltage gain of 9.5 V/V at V_{DD} of 2 V. Additionally, we compared the performance of our inverter with those of previously

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constructed inverters of similar architectures, and our results demonstrated comparable performance at low supply voltages ranging from 0.5 to 1.5 V. In this study, we confirmed the feasibility of replacing Si-based transistors with transistors based on 2D materials in a monolithic 3D circuit. This study highlights the potential of 2D materials for next-generation transistors and their application in advanced integrated circuit.

4. Experimental Section

Synthesis of Large-Area Monolayer MoS_2 Film: A large-scale MoS_2 film was prepared in a low-pressure CVD furnace equipped with a small inner tube. A small tube was loaded with molybdenum (VI) oxide (Alfa Aesar, 99.998%, and 50 mg) and supplied with Ar/O_2 gas (80/3 sccm). A silica crucible loaded with sulfur (Alfa Aesar, 99.999%, and 10 g) was put upstream of the substrate and supplied with Ar (150 sccm). Before the growth was commenced, sapphire wafers were first annealed at 1050 °C for 3 h to achieve an atomically flat surface. The temperatures of sulfur, molybdenum (VI) oxide, and substrate were 150, 550, and 900 °C, respectively. The full coverage of MoS_2 film was achieved after the 35–40 min of growth.

Synthesis of Monolayer WSe_2 Flakes: WSe_2 flakes were synthesized in a single-zone furnace equipped with a 2-inch quartz tube. Meanwhile, 0.3 g of WO_3 (99.999%; Alfa Aesar) and a sapphire substrate were placed in the middle of the furnace, whereas 0.4 g of Se powder (99.9%; Alfa Aesar) was placed in the upstream of the substrate. The distance between the substrate and WO_3 was about 2 cm and the distance between the substrate and Se was about18 cm. Before growth, the tube was degassed for 20 min by the oil pump. The temperature of the WO_3 and Se power were 840 and 260 °C, respectively. The growth lasted for 15 min under 100/40 sccm of Ar/H_2 gas flow.

Device Fabrication Processes—Transfer of Large-Area MoS_2 Film and WSe_2 Flakes: First, PMMA (Kayaku, PMMA 950 A4) was spin-coated on MoS_2/sapphire and annealed at 180 °C for 1 min. Subsequently, a rigid thermal release tape was attached to the PMMA/MoS_2/sapphire substrate. Next, the TRT/PMMA/MoS_2 stack was gently detached in 10 M KOH solution and then transferred to the target substrate and baked at 110 °C to release TRT. Finally, the PMMA was removed using hot acetone and isopropyl alcohol (IPA), respectively. WSe_2 was transferred onto the target substrate using a modified wet transfer method reported by the group.[35] First, the as-grown WSe_2 was first spin-coated with PMMA (Kayaku, PMMA 950 A4) at 800 rpm for 10 s and 2000 rpm for 30 s and then baked at 120 °C for 5 min. The PMMA/WSe_2 was separated by wet etching in KOH solution and transferred to a low-polarity solution for wrinkle-less transfer on the target substrate.

Device Fabrication Processes—Atomic Layer Deposition (ALD) of a HfO_2 Film: ALD of HfO_2 was performed at 250 °C using Tetrakis (ethylmethylamido) hafnium (TEMAHf) and O_2 as precursors. The pressure of the chamber was maintained at ≈7 Pa under 250 sccm of high-purity Ar gas. The pulse/purge times for Hf and O_2 precursors were 7000 ms/12 s and 3000 ms/12 s, respectively.

Measurements and Characterization: The morphologies of the as-grown MoS_2 and WSe_2 were examined via OM. The Raman and PL characterizations of as-prepared MoS_2 and WS_2 were performed using micro-Raman spectroscopy (HORIBA, LabRAM, and HR800) excited using a 532 nm laser. A JEOL JEM-200F electron microscope was used to observe the cross-sectional of the M3D integrated inverters. The electrical performance of as-fabricated M3D integrated inverters was evaluated using a semiconductor analyzer (Agilent, B1500A) at room temperature under an ambient atmosphere.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

M.J.L. and W.J.L. contributed equally to this work. M.J.L., W.J.L., C.H.S., J.H.C, and Y.L.C. conceived and coordinated the study. M.J.L, W.J.L, and J.H.C performed the experiments and data analysis. C.Z.C., C.X.H, and Y.L.Y performed some characterizations. All authors discussed the results and commented on the manuscript. M.J.L., W.J.L., C.H.S., J.H.C, and Y.L.C. co-wrote the paper. All the authors discussed the results and the manuscript.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

2D TMDs, chemical vapor deposition, MoS_2 , monolithic 3D, WSe_2

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